TRACELESS SU(2) CHARACTERS AND $\mathbb{Z}/4$ INSTANTON GRADINGS FOR TWO-BRIDGE AND $(3,n)$-TORUS KNOTS

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ABSTRACT. For the $(3,n)$-torus knots we determine the $\mathbb{Z}/4$ gradings of the reduced singular instanton chain complex through the double branched cover $\Sigma(2,3,n)$ rather than through an index computation: each irreducible flat connection on the cover has two traceless knot lifts of equal grading, so Daemi–Scaduto's theorem for the irreducible torus-knot complex transports to a grading split. For n odd this recovers the chain-rank distribution $(1+a,a,a,a)$, $a=-\sigma/4$, conjectured by Poudel–Saveliev and established for all torus knots by Daemi–Scaduto. Comparing that rank vector with rank $I^\natural = \|\Delta\|_1$ then determines the total rank of the framed differential in every residue class of n modulo 6: it vanishes for $n \equiv 1,2$ and equals one for $n \equiv 4,5$. The comparison uses no index theory, which is what carries it into the even classes, where $\Sigma(2,3,n)$ is not a homology sphere; $T(3,8)$ is a non-alternating knot with vanishing framed differential. Along the way we prove that every irreducible traceless SU(2) character of a two-bridge knot is binary-dihedral, with the traceless Riley polynomial in closed form, and that for the $(3,n)$-torus knots exactly $(\det -1)/2$ characters are dihedral, so for n odd none is; and we identify the first nonzero pillowcase differential, for $8_{19} = T(3,4)$, as a corner figure-eight bigon absent on two-bridge knots.

1. Introduction

1.1. The Atiyah–Floer conjecture and its knot version. Let Y be a closed oriented 3-manifold with a Heegaard splitting $Y = Y_0 \cup_\Sigma Y_1$. Two Floer theories attach to this picture. On the gauge-theoretic side is Floer's instanton homology $HF^{\mathrm{inst}}(Y)$, generated by flat SU(2) connections and with a differential counting anti-self-dual instantons on $Y \times \mathbb{R}$. On the symplectic side, the moduli space $M(\Sigma)$ of flat SU(2) connections on Σ is symplectic, the handlebodies determine Lagrangians $L_0, L_1 \subset M(\Sigma)$, and one forms the Lagrangian Floer homology $HF^{\mathrm{symp}}(L_0, L_1)$. The Atiyah–Floer conjecture [Ati88] asserts these are isomorphic (for a recent survey of the Lagrangian Floer theory in which the conjecture sits, see Auroux [Aur25]). The one general theorem is the mapping-torus case of Dostoglou–Salamon [DS94]; the Heegaard-splitting statement remains open, obstructed by the singularities of $M(\Sigma)$ and by figure-eight bubbling in the adiabatic limit.

There is a knot-theoretic analogue in which the pillowcase plays the role of $M(\Sigma)$. Let $K \subset S^3$ be a knot, and cut S^3 along a Conway sphere meeting K in four points into two 2-string tangles. A representation $\rho\colon \pi_1(S^3 \setminus K) \to \mathrm{SU}(2)$ is *traceless* if every meridian maps to a traceless element (conjugate to $\mathrm{diag}(i,-i)$); this is the boundary condition of Kronheimer–Mrowka's singular instanton theory [KM11a, KM11b]. Restricting traceless flat connections of each tangle to the four-punctured sphere and applying holonomy perturbations, one obtains immersed Lagrangians L_0, L_1 in the *pillowcase*

$$\mathcal{P} = (\mathbb{R}/2\pi\mathbb{Z})^2/\iota, \qquad \iota(\gamma,\theta) = (-\gamma,-\theta),$$

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the traceless SU(2) character variety of the four-punctured sphere: a two-sphere with four $\mathbb{Z}/2$ orbifold points [Lin92, HHK14]. The *knot Atiyah–Floer conjecture* of Hedden–Herald–Kirk and Cazassus–Herald–Kirk–Kotelskiy [HHK14, CHKK22] asserts that a bounding-cochain-deformed Lagrangian Floer homology of (L_0, L_1) is isomorphic, as a relatively $\mathbb{Z}/4$-graded group, to the reduced singular instanton knot homology $I^\natural(K)$. (A *bounding cochain* is a formal combination of self-intersection points of an immersed Lagrangian that deforms the Floer differential by higher-polygon counts — the standard device by which Lagrangian Floer theory absorbs disk bubbling [Aur25, CHKK22]; *figure-eight bubbling* is the degeneration of holomorphic strips, specific to immersed Lagrangians, that makes the correction necessary here [CHKK22].)

1.2. **Purpose and scope.** Two things here are new. The first is a determination of the $\mathbb{Z}/4$ grading distribution for the $(3,n)$-torus knots that runs through the double branched cover rather than through an index calculation: the two traceless knot lifts of a flat connection on $\Sigma(2,3,n)$ carry the same grading, so Daemi–Scaduto's theorem for the irreducible torus-knot complex transports to a grading split, and hence to the rank vector of the framed complex. The second is that comparing that rank vector with $\|\Delta\|_1$ determines the total rank of the framed differential in *all four* residue classes of n modulo 6, including the even classes, where the Fintushel–Stern index is unavailable because $\Sigma(2,3,n)$ is not a homology sphere. Both are collected in Theorem 1.3.

Around them the paper makes explicit, for two tractable infinite families, the representation-theoretic data on which the pillowcase construction rests: the traceless character variety (the generators), the $\mathbb{Z}/4$ spectral-flow gradings, and the differential. The same data is in active use beyond the pillowcase program: Dai–Mallick–Taniguchi have recently used an explicit analysis of the traceless character varieties of two-bridge knots to compute, in part, symmetry actions on their singular instanton complexes [DMT26].

Three explicit verifications are reported below: the exact-arithmetic check of Theorem 1.1, the counts of Proposition 4.1, and the gradings of Section 5. They are carried out by short computer programs, included as ancillary files with this preprint and also available at the public repository https://github.com/bwuebben/pillowcase-bounding-cochain. Each re-derives its output by an independent route; the first works in exact Gaussian-integer arithmetic.

1.3. **Results.** Section 2 records the pillowcase and the traceless characters as generators. Section 3 proves:

Theorem 1.1 (Traceless characters of two-bridge knots are dihedral). *Let $K = \mathfrak{b}(p,q)$ be the two-bridge knot of fraction p/q, so $p = \det K$. Every irreducible traceless* SU(2) *representation of $\pi_1(S^3 \setminus K)$ is binary-dihedral. There are exactly $(p-1)/2$ of them: writing ω for the meridian-pair angle (the angle between the axes of two chosen meridians, the conjugacy invariant of Section 2), they occur at $\cos\omega = \cos(2\pi k/p)$ for $k = 1, \ldots, (p-1)/2$, independent of q; equivalently the traceless Riley polynomial in $u = 2\cos\omega - 2$ is*

$$\phi_p(u) = \prod_{k=1}^{(p-1)/2} \Big(u + 4\sin^2 \tfrac{\pi k}{p}\Big),$$

monic of degree $(p-1)/2$, with integer coefficients, $\phi_p(0) = p = \det K$, and root-sum $-p$.

We then explain (Section 3.2) how Theorem 1.1 clarifies the Hedden–Herald–Kirk two-bridge isomorphism and localizes the obstruction, and we correct a naive expectation about the earring correspondence.

Section 4 computes the $(3,n)$-torus knot characters and proves:

Theorem 1.2 (Dihedral dichotomy for $(3,n)$-torus knots). *Let $n \geq 2$ and $\gcd(3,n) = 1$. The irreducible traceless* SU(2) *characters of $T(3,n)$ are in bijection with the traceless arcs of Section 4* *(the arcs of irreducible characters that contain a traceless point), one per arc. Exactly* $(\det -1)/2$ *of them are binary-dihedral: one if n is even* $(\det = 3)$ *and none if n is odd* $(\det = 1)$*. In particular, for n odd every irreducible traceless character of $T(3,n)$ is non-dihedral.*

Section 5 passes to the double branched cover $\Sigma(2,3,n)$ and proves the main result of the paper. Write $C_*(K)$ for the irreducible singular instanton complex and $\widetilde{C}(K) = C_*(K) \oplus C_{*-1}(K) \oplus \mathbb{Z}_{(0)}$ for the framed complex of the S-complex of [DS24b, Def. 3.21], whose homology is $I^\natural(K)$; this is not the framed instanton homology $I^\#$ of a closed three-manifold. Subscripts on $\mathbb{Z}$ denote gradings and superscripts denote ranks. Besides the differential d of C_*, the differential $\widetilde{\partial}$ of $\widetilde{C}$ has a map $v\colon C_* \to C_{*-2}$ and two maps $\delta_1\colon C_1 \to \mathbb{Z}$, $\delta_2\colon \mathbb{Z} \to C_{-2}$ to and from the reducible summand. For torus knots d vanishes, so all of the content of Theorem 1.3 lies in v, δ_1, δ_2.

Theorem 1.3 (Gradings and the framed differential for $(3,n)$-torus knots)**.** *Let* $\gcd(3,n) = 1$ *and put $a = -\sigma(T(3,n))/4$. For n odd, the irreducible flat* $\mathrm{SO}(3)$ *connections on $\Sigma(2,3,n)$ split evenly between the $\mathbb{Z}/4$ gradings $\mu = 1$ and $\mu = 3$, and*

$$\widetilde{C}(T(3,n)) = (1+a,\ a,\ a,\ a)$$

as a rank vector in gradings $0,1,2,3$. For every n coprime to 3, the total rank of the differential of $\widetilde{C}(T(3,n))$ is

$$\operatorname{rank}\widetilde{\partial} = \begin{cases} 0, & n \equiv 1,2 \pmod 6,\\ 1, & n \equiv 4,5 \pmod 6.\end{cases}$$

In particular $T(3,8)$ has vanishing framed differential although it is non-alternating.

Theorem 1.3 is Propositions 5.3 and 5.5 below. Section 6 reproduces the 8_{19} differential and identifies its geometric source. Section 7 states the genuinely open geometric direction.

1.4. **Strategy.** The arguments are short and largely independent, and we sketch them here.

Theorem 1.1 is a two-sided count. Irreducible binary-dihedral representations are automatically traceless, which produces at least $(p-1)/2$ traceless characters; the traceless specialization $s = i$ of Riley's polynomial has u-degree at most $(p-1)/2$, which produces at most that many. The bounds coincide, so the two sets are equal and the roots can be read off.

Theorem 1.2 turns on a trace obstruction. Writing $\pi_1(S^3 \setminus T(3,n)) = \langle x, y \mid x^3 = y^n\rangle$, the trace of $\rho(x)$ is ± 1, so $\rho(x)$ cannot lie in the reflection coset of $\mathrm{Pin}(2)$; irreducibility then forces $\rho(y)$ into that coset, where every element is traceless, and that is possible only when n is even. The converse is an explicit construction.

Theorem 1.3 has two halves, proved by different means. For the grading split we do not evaluate an index. Daemi–Scaduto compute the irreducible torus-knot complex: it is supported in gradings 1 and 3 with a generators in each and vanishing differential. The branched-cover correspondence sends each irreducible flat connection on $\Sigma(2,3,n)$ to a free pair of traceless knot lifts, and the two lifts carry the same grading; so the a cover connections must divide $a/2$ and $a/2$ between $\mu = 1$ and $\mu = 3$, and framing turns that into the rank vector. One step needs care: Daemi–Scaduto and Poudel–Saveliev normalize their gradings against different reference points, and the two differ by $\sigma \bmod 4$, which vanishes here because $\sigma = -4a$. Evaluating the Fintushel–Stern index and the equivariant ρ-invariant directly then serves as an independent check, for all odd $n \le 43$, rather than as the proof.

For the differential we compare two independently computed numbers. For n odd the chain rank is $1 + 4a$, from the first half; for n even it is instead $2N + 1$, since the irreducible complex then has one generator per traceless character and Proposition 4.1 counts those. The homology rank is pinned to $\|\Delta_{T(3,n)}\|_1$ by squeezing $I^\natural$ between Kronheimer–Mrowka's Alexander lower bound and the reduced Khovanov upper bound, the latter computed from Schütz's Bar–Natan chain decomposition for three-braids. Since $\dim H = \dim C - 2\,\mathrm{rank}\,\partial$, the difference of the two numbers *is* twice the differential rank. Nothing in that comparison uses the index, which is why it survives into the even classes, where $\Sigma(2,3,n)$ has nontrivial reducibles and the index formula does not exist.

1.5. **Three kinds of statement.** We are careful throughout to separate:

(i) results proved here (self-contained, though some are surely classical);
(ii) classical facts and standing theorems (attributed);
(iii) computations reproduced from [HHK18, PS17, Anv16] (attributed, and independently checked where possible).

Theorem 1.1 belongs to (i) but is elementary and essentially classical, and we say so again where it is proved. We claim no new theorem about the knot Atiyah–Floer conjecture itself: for two-bridge knots that is a theorem of Hedden–Herald–Kirk [HHK14, HHK18], and for the torus knots we treat the instanton side and the character-variety side, not the analytic identification of the two.

Acknowledgements. The literature computations reproduced here are due to Hedden–Herald–Kirk [HHK14, HHK18], Cazassus–Herald–Kirk–Kotelskiy [CHKK22], Poudel–Saveliev [PS17], and Anvari [Anv16]; we have tried to attribute each precisely.

2. The pillowcase and traceless characters

We work with SU(2) as the unit quaternions. An element is traceless iff it is a purely imaginary unit quaternion (eigenvalues $\pm i$, squaring to -1); as a rotation it is a rotation by π. The traceless elements form a 2-sphere $S \subset \mathrm{Im}\,\mathbb{H} \cong \mathbb{R}^3$.

Proposition 2.1 (Classical; cf. [Lin92, HHK14])**.** *The traceless* SU(2) *character variety of the four-punctured sphere is the pillowcase $\mathcal{P} = (\mathbb{R}/2\pi\mathbb{Z})^2/\iota$, with four $\mathbb{Z}/2$ orbifold points at $(\gamma,\theta) \in \{0,\pi\}^2$ and the symplectic form descending from $d\gamma \wedge d\theta$.*

Sketch. Write π_1 of the four-punctured sphere as $\langle X_1, \dots, X_4 \mid X_1X_2X_3X_4 = 1\rangle$ with each $X_i \in S$. Since purely imaginary units satisfy $X_i^{-1} = -X_i$, the relation gives $X_1X_2 = X_4X_3 =: A$. For pure imaginary units, $X_iX_j = -\langle X_i, X_j\rangle + X_i \times X_j$, so when $A \neq \pm 1$ its axis is well defined; equating the two expressions for A forces the four X_i to be coplanar, orthogonal to axis(A). Placing that axis along k and writing $X_m = \cos\beta_m\, i + \sin\beta_m\, j$, the relation becomes $\beta_2 - \beta_1 = \beta_3 - \beta_4$; gauge-fixing $\beta_1 = 0$ and setting $\gamma = \beta_2$, $\theta = \beta_4$ leaves two parameters, and the residual conjugation by the π-rotation about i acts by $(\gamma,\theta) \mapsto (-\gamma,-\theta)$. The four fixed points are the reducible (collinear) characters. □

For a knot K with a two-tangle decomposition, a representation of $\pi_1(S^3 \setminus K)$ is a compatible pair of tangle representations, i.e. a point of $L_0 \cap L_1$. Thus the intersection points are exactly the traceless representations of the knot group, and (after perturbation) the generators of the singular instanton chain complex [KM11a, KM11b]. For the two-bridge knots of Section 3, an irreducible traceless representation is determined up to conjugacy by the single invariant $\cos\omega = \langle a, b\rangle$, the cosine of the angle between the axes of two chosen meridians $a, b \in S$.

3. Two-bridge knots

3.1. **Traceless characters are dihedral.** Recall Riley's parametrization [Ril84]: for a two-bridge knot the nonabelian $\mathrm{SL}_2(\mathbb{C})$ representations with meridian eigenvalue s are

$$\rho(a) = \begin{pmatrix} s & 1 \\ 0 & s^{-1} \end{pmatrix}, \qquad \rho(b) = \begin{pmatrix} s & 0 \\ -u & s^{-1} \end{pmatrix},$$

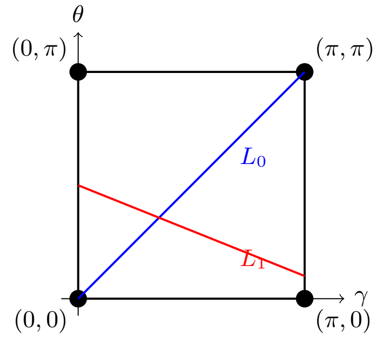

FIGURE 1. The pillowcase $\mathcal{P} = (\mathbb{R}/2\pi\mathbb{Z})^2/\iota$, a 2-sphere with four $\mathbb{Z}/2$ orbifold (corner) points, shown on the square $[0,\pi]^2$. For a two-bridge knot the two tangle Lagrangians are straight arcs of rational slope (Section 3.2); being straight they meet transversally and bound no bigon. The earring correspondence bends such an arc into a figure-eight at each corner (Remark 3.2).

cut out by a polynomial $\Phi_{p,q}(s,u)$ of degree $(p-1)/2$ in u; the traceless locus is $s = i$. On that locus $\operatorname{tr}\rho(ab) = 2-u$, so Riley's parameter is $u = 2 - \operatorname{tr}\rho(ab) = 2\cos\omega - 2$ in the meridian-pair angle ω of Section 2. Write ϕ_p for the monic normalization of $\Phi_{p,q}(i,\cdot)$; Theorem 1.1 shows it is independent of q.

Proof of Theorem 1.1. A binary-dihedral representation sends meridians into the reflection coset $e^{i\phi}j$ of $\mathrm{Pin}(2) = S^1 \cup jS^1$, whose elements are traceless; hence every irreducible dihedral representation is traceless, so there are *at least* $(p-1)/2$ irreducible traceless characters (the number of irreducible metabelian representations of a knot with $|H_1(\Sigma_2)| = p$ odd). On the other hand the traceless Riley polynomial $\Phi_{p,q}(i,u)$ has u-degree at most $(p-1)/2$, so there are *at most* $(p-1)/2$. The bounds coincide, so every irreducible traceless character is one of the dihedral ones. A dihedral character with $\rho(ab)$ of rotation angle $4\pi k/p$ has $\cos\omega = \cos(2\pi k/p)$; distinct $k \in \{1,\dots,(p-1)/2\}$ give distinct characters, and these values depend only on p. Finally $\phi_p(u) = \prod_k(u-u_k)$ with $u_k = 2\cos(2\pi k/p) - 2 = -4\sin^2(\pi k/p)$; its constant term is $\prod_k 4\sin^2(\pi k/p) = p$ and its root-sum is $\sum_k(2\cos(2\pi k/p)-2) = -1-(p-1) = -p$. The Galois group of the pth cyclotomic field permutes the set of roots $\{u_k\}$, so the monic product ϕ_p has rational coefficients; because its roots are algebraic integers, those coefficients are algebraic integers as well, and hence lie in $\mathbb{Z}$. $\square$

Remark 3.1. Small cases: $\phi_3 = u+3$, $\phi_5 = u^2+5u+5$, $\phi_7 = u^3+7u^2+14u+7$, $\phi_9 = u^4+9u^3+27u^2+30u+9$. For p prime, ϕ_p is, after $u = 2c-2$, the minimal polynomial of $2\cos(2\pi/p)$, irreducible of degree $(p-1)/2$. Theorem 1.1 is elementary and is essentially classical (the traceless/metabelian coincidence for two-bridge knots is implicit in [Ril84]); we include the proof because it fixes the interior generators exactly and q-independently, which is what the pillowcase picture needs. The statement was verified by exact Gaussian-integer computation of the Riley word for all $p \le 9$, including the two determinant-9 knots $9_1 = \mathfrak{b}(9,1)$ and the hyperbolic $6_1 = \mathfrak{b}(9,2)$, which produce the identical root set $\{-4\sin^2(\pi k/9)\}$.

3.2. **The pillowcase consequence and the earring.** The character variety of a rational tangle of slope r is a straight arc of slope r in the pillowcase, and the double branched cover of the four-punctured sphere is a torus T^2 on which the tangles lift to straight lines [Lin92, HHK14, CHKK22]. For $\mathfrak{b}(p,q)$ the two lifted lines have classes $(0,1)$ and (p,q); they meet in $|\det\left(\begin{smallmatrix}0&1\\p&q\end{smallmatrix}\right)| = p$ points, and gluing the two solid tori recovers $\Sigma_2(K) = L(p,q)$. Being straight lines, they bound no immersed bigon.

Two things follow, and one warning. First, the naive count on $\mathcal{P}$ itself is $(p+1)/2$: the $(p-1)/2$ interior dihedral characters of Theorem 1.1 together with the single abelian character at a corner. The passage to the full det $= p$ requires the *earring*: the reduction defining $I^\natural$ adds to the knot a small linking circle carrying a fixed holonomy, which removes the reducibles and replaces each tangle Lagrangian by its image under the earring correspondence $E \subset \mathcal{P}^- \times \mathcal{P}$ [KM11a, CHKK22, HK24]. Second, and structurally: the differential and the bounding-cochain deformation are assembled from holomorphic bigons and discs, of which straight lines have none. Thus on the two-bridge family the figure-eight bubbling that obstructs the general conjecture is inert.

This is consistent with the known outcome: Hedden–Herald–Kirk [HHK14, HHK18] prove that for every two-bridge knot the $\mathbb{Z}/4$ gradings agree and the differentials vanish on both sides, so pillowcase homology is isomorphic to $I^\natural$ with zero bounding cochain, of rank det. We emphasize this is *their* theorem; Theorem 1.1 only clarifies why the interior generators are exactly the dihedral characters and why the obstruction is corner-localized.

Remark 3.2 (Correcting a naive expectation)**.** It is tempting to hope that composing the straight Lagrangians with the Cazassus–Herald–Kirk–Kotelskiy earring correspondence $E \subset \mathcal{P}^- \times \mathcal{P}$ [CHKK22], where $\mathcal{P}^-$ is $\mathcal{P}$ with the sign of its symplectic form reversed, preserves straightness. It does not: Herald–Kirk [HK24] prove, at the regular-homotopy level, that E sends an arc-type Lagrangian to a genuine figure-eight curve with one self-crossing per corner, while doubling loop-type curves that avoid the corners. Thus $E \circ L_1$ is immersed, and all of its new geometry is confined to the four corners. For two-bridge knots the net effect is nonetheless trivial (zero bounding cochain suffices). The rigorously established nonzero examples, the $(4,5)$-torus knot [CHKK22] and the pretzel knot $P(-2,3,5)$ [Smi24], are both non-alternating; the companion paper [Wue26] reports additional finite polygon experiments for which the full Maurer–Cartan equation has not been established.

4. The $(3,n)$-torus knots

4.1. **The representation arcs.** Throughout this section $n \geq 2$ and $\gcd(3,n) = 1$. Write $\pi_1(S^3 \setminus T(3,n)) = \langle x, y \mid x^3 = y^n \rangle$. The element $z = x^3 = y^n$ is central, so any irreducible SU(2) representation sends $z \mapsto \pm 1$ and

$$\rho(x) = e^{\alpha\hat{u}}, \quad \rho(y) = e^{\beta\hat{v}}, \qquad \alpha = \tfrac{\pi\ell_1}{3}, \quad \beta = \tfrac{\pi\ell_2}{n},$$

with $\hat{u}, \hat{v}$ unit imaginary quaternions, $\ell_1 \in \{1,2\}$, $0 < \ell_2 < n$, and the central-sign match $\ell_1 \equiv \ell_2 \pmod 2$. The angle $\psi = \angle(\hat{u},\hat{v}) \in (0,\pi)$ is the only remaining modulus (we write ψ, not θ, to avoid collision with the pillowcase coordinate of Section 2); each such pair (ℓ_1,ℓ_2) is a one-parameter arc of irreducible characters (Klassen [Kla91]), and we call these the *representation arcs*. Call an arc a *traceless arc* if it contains a traceless character; the criterion, derived in the proof of Proposition 4.1 below, is $|\cot(b\beta)| < \sqrt{3}$.

4.2. **The traceless locus.** The meridian is $\mu = x^a y^b$ with $an + 3b = 1$; it is well defined modulo the center, and tracelessness is insensitive to that ambiguity (multiplying μ by z negates $\rho(\mu)$, preserving trace 0). A direct quaternion computation gives

$$\tfrac{1}{2}\operatorname{tr}\rho(\mu) = \operatorname{Re}\rho(\mu) = \cos(a\alpha)\cos(b\beta) - \sin(a\alpha)\sin(b\beta)\cos\psi,$$

which is affine in $\cos\psi$. Hence each arc contains exactly one traceless character, at

$$\cos\psi = \cot(a\alpha)\,\cot(b\beta), \tag{1}$$

provided the right side lies in $(-1,1)$ (otherwise the arc carries none; the boundary case never occurs, as the proof below shows). We verified in each case that the resulting ρ satisfies $\rho(x)^3 = \rho(y)^n = (-1)^{\ell_1}$ and $\operatorname{tr}\rho(\mu) = 0$ exactly; both central signs occur, and for n odd the traceless characters split evenly between the two sheets (see the proof of Proposition 4.1).

Proposition 4.1 (Count)**.** *The number of irreducible traceless* $\mathrm{SU}(2)$ *characters of* $T(3,n)$ *is*

$$N(3,n) = \begin{cases} 2(n-1)/3, & n \equiv 1 \pmod 6, \\ (2n-1)/3, & n \equiv 2 \pmod 6, \\ (2n+1)/3, & n \equiv 4 \pmod 6, \\ 2(n+1)/3, & n \equiv 5 \pmod 6, \end{cases}$$

together with one abelian traceless character; thus $N(3,n) \sim 2n/3$.

Proof. From $an + 3b = 1$ we get $\gcd(a,3) = \gcd(b,n) = 1$. Since $\ell_1 \in \{1,2\}$ and $3 \nmid a\ell_1$, the first cotangent in (1) always has $|\cot(a\alpha)| = \cot(\pi/3) = 1/\sqrt{3}$; so an arc is a traceless arc iff $|\cot(b\beta)| < \sqrt{3}$, iff $\pi b\ell_2/n$ lies in $(\pi/6, 5\pi/6)$ modulo π, iff the residue $m = b\ell_2 \bmod n$ lies in the open interval $(n/6,\, 5n/6)$. Equality never occurs: it would force $6 \mid n$, impossible with $\gcd(3,n) = 1$, so the boundary case $\cos\psi = \pm 1$ does not arise. The parity constraint assigns to each $\ell_2 \in \{1,\dots,n-1\}$ exactly one ℓ_1 ($\ell_1 = 1$ for ℓ_2 odd, $\ell_1 = 2$ for ℓ_2 even), and $\ell_2 \mapsto m = b\ell_2 \bmod n$ is a bijection of $\{1,\dots,n-1\}$; hence

$$N(3,n) = \#\{m \in \mathbb{Z} :\ n/6 < m < 5n/6\} = \lfloor 5n/6 \rfloor - \lfloor n/6 \rfloor,$$

which evaluates, in the four residues $n \equiv 1,2,4,5 \pmod 6$, to the stated cases. Distinct arcs give distinct characters, since $\operatorname{tr}\rho(y) = 2\cos(\pi\ell_2/n)$ separates them, and the abelian traceless character (every meridian to a fixed traceless element) is unique up to conjugacy. Finally, for n odd the involution $\ell_2 \mapsto n - \ell_2$ flips the parity of ℓ_2 (hence exchanges the sheets $\ell_1 = 1 \leftrightarrow 2$) and preserves admissibility, since $m \mapsto n - m$ preserves the interval; being fixed-point-free, it splits the $N(3,n)$ traceless characters evenly between the two central-sign sheets, as claimed above. □

The values $N(3,n)$ for $n = 2,4,5,7,8,10,11,13$ are $1,3,4,4,5,7,8,8$. As a check, the formula was also verified by direct enumeration of the traceless arcs for all $n \le 25$.

4.3. **The dihedral dichotomy.**

Proof of Theorem 1.2. Conjugate a binary-dihedral image into $\mathrm{Pin}(2) = S^1 \cup jS^1$. Since $\operatorname{tr}\rho(x) = 2\cos(\pi\ell_1/3) = \pm 1$, the element $\rho(x)$ cannot lie in the reflection coset jS^1; it therefore lies in S^1. Irreducibility forces $\rho(y)$ to lie in jS^1, whose elements are traceless. Thus $\beta = \pi/2$, equivalently $\ell_2 = n/2$, so n must be even.

Conversely, suppose n is even and take $\ell_2 = n/2$. The parity constraint determines a unique $\ell_1 \in \{1,2\}$. In the Bézout relation $an + 3b = 1$, the integer b is odd, and hence $\cot(b\beta) = \cot(b\pi/2) = 0$. Equation (1) gives $\cos\psi = 0$. Relative to the circle S^1 containing $\rho(x)$, this places $\rho(y)$ in jS^1; the resulting traceless character is irreducible and binary-dihedral. Hence there is exactly one such character for even n and none for odd n. This equals $(\det T(3,n) - 1)/2$, since $\det T(3,n) = 3$ for even n and 1 for odd n. □

Theorem 1.2 is the exact opposite of Theorem 1.1: two-bridge knots are entirely dihedral, the $(3,n)$-torus knots with n odd entirely non-dihedral. The non-dihedral characters are precisely those the pillowcase construction represents by non-straight Lagrangians, on which the earring produces genuine figure-eight curves; this is why the family is a natural first test beyond the metabelian world.

Remark 4.2. There is a visible symmetry: $(\ell_1, \ell_2) \mapsto (3-\ell_1,\, n-\ell_2)$ preserves ψ in (1) (both cotangents change sign), and it respects the parity constraint $\ell_1 \equiv \ell_2 \pmod 2$ exactly when n is odd; in that case it pairs the two central-sign sheets $\rho(x)^3 = \pm 1$ at equal meridian angle. This is the even split in the proof of Proposition 4.1.

5. The $\mathbb{Z}/4$ gradings via the double branched cover

The double branched cover of $T(3,n)$ is the Brieskorn sphere $\Sigma(2,3,n)$. For odd n it is an integral homology sphere; its irreducible flat SO(3) connections have rotation numbers $(m_1, m_2, m_3) = (1, m_2, m_3)$ indexing the three singular fibers, with m_2, m_3 even, $0 < m_2 < 3$, $0 < m_3 < n$. Since 2 is the only even value in $(0,3)$, necessarily $m_2 = 2$, and m_3 ranges over the even integers in $(0,n)$ meeting the spherical-triangle inequalities on the angles $(\pi/2, 2\pi/3, \pi m_3/n)$. (These m_i index the branched cover and are not the (ℓ_1, ℓ_2) of Section 4.) Fintushel–Stern's count gives $a = -\sigma(T(3,n))/4$ such connections [FS90]. They are equivariant, and each has two traceless knot lifts [PS17, §7.4]; equivalently, these are the two-element fibers in the branched-cover correspondence [DS24b, §9.1]. Consequently

$$N(3,n) = 2a \qquad (n \text{ odd}), \qquad \sigma(T(3,n)) = -2N(3,n), \tag{2}$$

an identity independently checked by the triangle-inequality count for every odd $n \le 43$. Thus the character count of Proposition 4.1 determines the signature and, with the theorem below, the graded chain group. Throughout this section a relatively $\mathbb{Z}/4$-graded group is recorded by its rank vector (r_0, r_1, r_2, r_3), the ranks in gradings $0,1,2,3$; unqualified homology ranks mean free ranks, equivalently dimensions after tensoring with $\mathbb{Q}$.

5.1. **The grading formula.** The $\mathbb{Z}/4$ grading is the spectral flow of the extended Hessian mod 4, relative to the trivial connection Θ (a single generator, in grading σ mod 4); every irreducible contributes four generators, two in grading μ and two in $\mu+1$ [KM11b, PS17]. The knot grading is

$$\mu(\alpha) = \tfrac{1}{2}\,\mathrm{gr}(\alpha) + \tfrac{1}{4}\big(1 - \rho_{\mathrm{Ad}\,\alpha}\big) \pmod 4, \tag{3}$$

where, for $\Sigma(2,p,q)$ with $p=3$, $q=n$, the Fintushel–Stern index [FS90, Anv16] and the equivariant ρ-invariant [Anv16] are

$$\mathrm{gr}(\alpha) = \frac{e^2}{pq} + \frac{2}{p}\sum_{k=1}^{p-1} \cot\tfrac{\pi k}{p}\cot\tfrac{\pi b_2 k}{p}\sin^2\tfrac{\pi k m_2}{p} + \frac{2}{q}\sum_{k=1}^{q-1} \cot\tfrac{\pi k}{q}\cot\tfrac{\pi b_3 k}{q}\sin^2\tfrac{\pi k m_3}{q}, \tag{4}$$

$$\rho_{\mathrm{Ad}\,\alpha} = 1 - \frac{2}{pq} - \frac{4}{p}\sum_{k=1}^{p-1} \cot\tfrac{\pi k}{p}\tan\tfrac{\pi b_2 k}{p}\cos^2\tfrac{\pi k m_2}{p} - \frac{4}{q}\sum_{k=1}^{q-1} \cot\tfrac{\pi k}{q}\tan\tfrac{\pi b_3 k}{q}\cos^2\tfrac{\pi k m_3}{q}, \tag{5}$$

with $e = pq\,m_1 + 2q\,m_2 + 2p\,m_3$ and b_2, b_3 the even Seifert invariants ($pqb_1 + 2qb_2 + 2pb_3 = 1$); since the cotangents see them only mod p, q, one may take $b_2 \equiv (2q)^{-1}$, $b_3 \equiv (2p)^{-1}$.

Example 5.1. For $\Sigma(2,3,7)$ the two connections $(1,2,2)$ and $(1,2,4)$ give $\mathrm{gr} = 175 \equiv 7$ and $\mathrm{gr} = 251 \equiv 3 \pmod 8$, $\rho = 3$ in both cases, and $\mu = 87 \equiv 3$ and $\mu = 125 \equiv 1 \pmod 4$. This agrees with Anvari's Example 6.1 [Anv16] at the level of the unreduced values, confirming our evaluation of (4)–(5).

5.2. **The homology ranks.** The rank of $I^\natural$ is pinned independently of any grading calculation, by a two-sided bound. This is used for both parities below.

Lemma 5.2 (Homology ranks)**.** *For every $n \ge 2$ coprime to 3,*

$$\mathrm{rank}_{\mathbb{Q}}\, I^\natural(T(3,n);\mathbb{Q}) = \|\Delta_{T(3,n)}\|_1 = \begin{cases} 4k+1, & n = 3k+1,\\ 4k+3, & n = 3k+2.\end{cases}$$

Proof. Schütz works over the Bar–Natan ring $\mathbb{Z}[X,h]/(X^2-Xh)$ and writes A for its q-shifted rank-one free module and $A(j)$ for the two-term complex with differential $(2X-h)^j$ [Sch25, Def. 3.2]; the reduced Khovanov complex is obtained by tensoring the Bar–Natan complex with $\mathbb{Z}$, where X and h act by zero. His chain decompositions give, with grading shifts suppressed,

$$C_{BN}(T(3,3k+1)) \simeq A \oplus A(1)^k \oplus A(2)^k,$$
$$C_{BN}(T(3,3k+2)) \simeq A \oplus A(1)^{k+1} \oplus A(2)^k$$

[Sch25, §3 and Thm. 5.3(2),(4)]. Each $A(j)$ is a two-term complex with differential $(2X-h)^j$, so after reduced specialization it is a zero-differential complex on two generators. Consequently

$$\operatorname{rank}_{\mathbb{Q}} \widetilde{Kh}(T(3,n);\mathbb{Q}) = \begin{cases} 4k+1, & n = 3k+1, \\ 4k+3, & n = 3k+2. \end{cases} \tag{6}$$

Up to multiplication by a monomial, the Alexander polynomial is

$$\Delta_{T(3,n)}(t) = \frac{1+t^n+t^{2n}}{1+t+t^2}.$$

Below degree n its coefficients repeat $1,-1,0$, and the quotient is reciprocal of degree $2n-2$. The central coefficient, in degree $n-1$, has absolute value one. If $n=3k+1$, the strictly lower half has $2k$ nonzero coefficients; if $n=3k+2$, it has $2k+1$. Reflection across the center therefore gives coefficient norm $4k+1$ and $4k+3$, respectively, exactly the right side of (6). Since $KHI(K)$ is Alexander-graded with graded Euler characteristic $\Delta_K(t)$, one has $\operatorname{rank} KHI(K) \geq \|\Delta_K\|_1$; the reduced Khovanov-to-instanton spectral sequence gives the upper bound (the spectral sequence starts from the mirror, which has the same rational Khovanov rank); together with $I^\natural(K;\mathbb{Q}) \cong KHI(K;\mathbb{Q})$, this yields

$$\|\Delta_{T(3,n)}\|_1 \leq \operatorname{rank}_{\mathbb{Q}} I^\natural(T(3,n);\mathbb{Q}) \leq \operatorname{rank}_{\mathbb{Q}} \widetilde{Kh}(T(3,n);\mathbb{Q}) = \|\Delta_{T(3,n)}\|_1$$

[KM10, Thm. 1.1][KM11b, Props. 1.2 and 1.4]. Hence equality holds throughout. □

5.3. **The grading split for n odd.** The family-wide grading distribution follows from Daemi–Scaduto's irreducible torus-knot theorem. Direct evaluation of (4)–(5) supplies an independent finite check.

Proposition 5.3. *Let n be odd and coprime to 3, and set $a = -\sigma(T(3,n))/4 = N(3,n)/2$ (an even integer). Under the grading (3), the irreducible connections of $\Sigma(2,3,n)$ split evenly: $a/2$ have $\mu = 1$ and $a/2$ have $\mu = 3$ (none has $\mu = 0$ or 2). Consequently the framed complex $\widetilde{C}$ has rank vector*

$$\widetilde{C}(T(3,n)) = (1+a,\ a,\ a,\ a), \qquad \chi = +1,$$

with the trivial generator Θ in grading 0. If $\Delta_{T(3,n)}(t) = \sum_j c_j t^j$ and $\|\Delta_{T(3,n)}\|_1 = \sum_j |c_j|$, then the homology *$I^\natural$ satisfies*

$$\operatorname{rank} I^\natural(T(3,n)) = \|\Delta_{T(3,n)}\|_1 = \begin{cases} 1+4a, & n \equiv 1 \pmod 6 \quad (\widetilde{\partial} = 0), \\ 4a-1 = (1+4a)-2, & n \equiv 5 \pmod 6 \quad (\operatorname{rank} \widetilde{\partial} = 1), \end{cases}$$

where $\widetilde{\partial}$ is the differential of $\widetilde{C}$ and its rank means total rational rank. Thus $\widetilde{\partial}$ vanishes for $n \equiv 1$ and has total rank one for $n \equiv 5$.

Remark 5.4 (What the rank vector recovers, and what it does not)**.** The rank vector $(1+a,a,a,a)$ is the chain-rank distribution conjectured by Poudel–Saveliev [PS17, §7.4]. That conjecture was resolved for all torus knots by Daemi–Scaduto [DS24, Cor. 3], whose theorem the proof below takes as an input; see [DS24b, §9.5] for the equivalence of the two formulations. What is added here is the branched-cover route to the distribution for this family, and the differential ranks that follow from it. The comparison is moreover at the level of $\widetilde{C}$, which has the same homology as $C^\natural$ [DS24b, Thm. 8.9] though not the same chain ranks.

The smallest case with $\widetilde{\partial} \neq 0$ is $T(3,5) = P(-2,3,5) = 10_{124}$, whose homology has rank 7 rather than the chain rank 9. On the pillowcase side Smith proves independently that the relevant bounding cochain is nonzero [Smi24]. The two facts are consistent, and we do not identify that cochain with $\widetilde{\partial}$.

Proof. Let $C_*(K)$ denote the irreducible singular instanton complex and $I_*(K) = H_*(C_*(K))$ its homology. Daemi–Scaduto prove that for every torus knot K

$$I_*(K) \cong \mathbb{Z}_{(1)}^{\lceil -\sigma(K)/4 \rceil} \oplus \mathbb{Z}_{(3)}^{\lfloor -\sigma(K)/4 \rfloor};$$

[DS24, Cor. 3]; the torus-knot complex has total rank $-\sigma(K)/2$ and is supported in gradings 1 and 3 [DS24b, §9.5], so its differential vanishes. For the odd knots under consideration, $-\sigma/4 = a$, and hence

$$C_*(T(3,n)) \cong \mathbb{Z}_{(1)}^a \oplus \mathbb{Z}_{(3)}^a.$$

The *branched-cover correspondence* relates the two sides. Flat SO(3) connections on the double branched cover $\Sigma(2,3,n)$ that are equivariant for the deck involution correspond to traceless SU(2) representations of the knot group, and the deck symmetry descends to an involution ι on the knot side, the *flip*. Over an irreducible cover connection β the fiber is a free ι-orbit $\{\alpha, \iota\alpha\}$ of two irreducible knot classes [DS24b, §9.1]. (Two disambiguations: this ι is not the pillowcase involution of Section 2, and α, β denote connections and classes here, not the angles $\pi\ell_1/3$, $\pi\ell_2/n$ of Section 4.) In the Poudel–Saveliev model the same fiber appears as a pair of circles of critical points, interchanged by the flip [PS17, Prop. 4.1(c), read as in §6.2]; nondegeneracy holds for these torus-knot classes [PS17, §7.4], and the two circles carry the same μ-grading [PS17, Lem. 6.4 and the paragraph following it].

The two sides grade their generators against different reference points, so they must be aligned before the counts can be compared. Daemi–Scaduto grade an irreducible α by the index of the singular cylinder joining it to the reducible, normalized so that the reducible sits in degree zero [DS24b, (2.18)]; Poudel–Saveliev grade the pair of circles over β by an equivariant index carrying an additive sign term [PS17, Prop. 6.5]. The two normalizations differ by exactly the degree of the comparison isomorphism, which is $\sigma(K)$ mod 4 [DS24b, Thm. 8.9]. Hence the generator-level dictionary

$$\mu(\beta) = \mathrm{gr}_{DS}(\alpha) + \sigma(T(3,n)) \pmod 4,$$

where gr_{DS} is the Daemi–Scaduto grading on the knot complex, not the Fintushel–Stern index gr of (4). Because $\sigma = -4a$, the shift vanishes modulo 4. If ν_j denotes the number of cover connections of degree j, the two equal-degree lifts of each β and the displayed ranks of C_* give

$$2\nu_1 = a, \qquad 2\nu_3 = a, \qquad \nu_0 = \nu_2 = 0.$$

Thus $a/2$ cover connections have two lifts of degree 1, and $a/2$ have two lifts of degree 3. In particular a is even; explicitly $a = (n-1)/3$ for $n \equiv 1 \pmod 6$ and $a = (n+1)/3$ for $n \equiv 5 \pmod 6$.

The framed complex $\widetilde{C}_* = C_* \oplus C_{*-1} \oplus \mathbb{Z}_{(0)}$ of Section 1.3 carries the differential of an S-complex [DS24b, Def. 3.21]: besides the differential d of C_* it has a map $v\colon C_* \to C_{*-2}$ and two maps $\delta_1\colon C_1 \to \mathbb{Z}$, $\delta_2\colon \mathbb{Z} \to C_{-2}$ to and from the reducible summand. The $\mu = 1$ cover connections contribute two generators in each of degrees $1, 2$, while the $\mu = 3$ connections contribute two in each of degrees $3, 0$; the reducible Θ adds one in degree $\sigma \equiv 0$. This directly gives the rank vector $(1+a, a, a, a)$. Daemi–Scaduto give a graded chain-homotopy equivalence from the reduced singular instanton complex to $\widetilde{C}$, and hence its homology computes $I^\natural$ [DS24b, Thm. 8.9]. The vanishing used above concerns only d; the maps v, δ_1, δ_2 need not vanish, which is why $\widetilde{\partial}$ can be nonzero.

It remains to determine the homology rank, and Lemma 5.2 supplies it. For $n \equiv 1 \pmod 6$ the common rank is $1+4a$, while for $n \equiv 5 \pmod 6$ it is $4a-1$. Finally, for any finite-dimensional complex, $\dim H = \dim C - 2\operatorname{rank}\partial$; applying this to $\widetilde{C}$, of chain rank $1+4a$ gives total differential rank zero and one, respectively. Direct evaluation of (4)–(5) for all odd $n \le 43$ independently verifies the grading split. Anvari's calculation proves the $n \equiv 1 \pmod 6$ subfamily directly [Anv16, Ex. 6.2]. ☐

Explicitly $a = (n-1)/3$ for $n \equiv 1$ and $a = (n+1)/3$ for $n \equiv 5 \pmod 6$; either way $\operatorname{rank} I^\natural \sim 2N(3,n)$ grows while $\det = 1$ stays fixed, so the rank far exceeds the determinant. Here $\chi(I^\natural(K)) = +1$ for *every* knot [PS17, Thm 8.1], carried by the single trivial generator, so χ coincides with $\det = 1$ only by accident; in general $\chi \neq \det$, as the case 8_{19} below ($\chi = 1$, $\det = 3$) shows.

n	5	7	11	13	17	19	23	25
n mod 6	5	1	5	1	5	1	5	1
$N(3,n) = 2a$	4	4	8	8	12	12	16	16
$a = -\sigma/4$	2	2	4	4	6	6	8	8
chain rank $1+4a$	9	9	17	17	25	25	33	33
rank $I^\natural$	**7**	9	**15**	17	**23**	25	**31**	33

TABLE 1. For $T(3,n)$, n odd coprime to 3: the traceless character count $N = 2a$, the standard chain rank $\dim \widetilde{C} = 1+4a$ (Proposition 5.3), and the homology rank $\operatorname{rank} I^\natural = \|\Delta_{T(3,n)}\|_1$. The two coincide for $n \equiv 1 \pmod 6$ (differential zero) but differ by 2 for $n \equiv 5 \pmod 6$ (bold; $\operatorname{rank}\widetilde{\partial} = 1$). $T(3,5) = P(-2,3,5) = 10_{124}$ is the smallest nonzero case; every displayed homology rank follows from Lemma 5.2. The chain-group rank vector for every column is $(1+a, a, a, a)$.

5.4. **The even residue classes.**

Proposition 5.5 (Even residue classes)**.** *Let n be even and coprime to 3, and write $N = N(3,n)$. The framed complex $\widetilde{C}$ has rank $2N+1$. Its rational homology rank and the total rank of its differential are*

n	N	$\operatorname{rank}_{\mathbb{Q}} I^\natural(T(3,n);\mathbb{Q})$	$\operatorname{rank}_{\mathbb{Q}} \partial$
$6m+2$	$4m+1$	$8m+3 = 2N+1$	0
$6m+4$	$4m+3$	$8m+5 = 2N-1$	1.

In particular, $\widetilde{C}$ has zero differential for every $n \equiv 2 \pmod 6$ and a nonzero differential of total rank one for every $n \equiv 4 \pmod 6$.

Proof. For torus knots, the irreducible complex has one generator for every irreducible traceless character and total rank N [DS24b, §9.5]; the framed construction $C_* \oplus C_{*-1} \oplus \mathbb{Z}_{(0)}$ therefore has rank $2N+1$. This is also the minimal complex produced from the torus-knot representation variety by Hedden–Herald–Kirk [HHK14, §§5.1 and 11]. Proposition 4.1 gives $N = 4m+1$ for $n = 6m+2$ and $N = 4m+3$ for $n = 6m+4$. Lemma 5.2 gives homology rank $8m+3$ and $8m+5$, respectively. The identity $\dim H = \dim C - 2\operatorname{rank}\partial$ applied to $\widetilde{C}$ gives the two asserted differential ranks. ☐

For example, $T(3,8)$ has $N=5$, chain rank 11, and homology rank 11, so its differential is zero, although $T(3,8)$ is not alternating. The vanishing is systematic rather than accidental: it happens for every $n \equiv 2 \pmod 6$, while the rank-one cases are exactly $n \equiv 4 \pmod 6$.

The Fintushel–Stern grading is special to n odd, where $\Sigma(2,3,n)$ is a homology sphere; for n even $b_3 \equiv (2p)^{-1} \bmod n$ does not exist. Proposition 5.5 replaces that index calculation by a rank argument: $\widetilde{\partial}$ vanishes for $n \equiv 2 \pmod 6$ and is nonzero for $n \equiv 4 \pmod 6$. The smallest nonzero instance is $8_{19} = T(3,4)$. In this case the corresponding pillowcase calculation is also known: its bounding cochain vanishes and its differential is a single bigon. We now turn to that calculation.

6. A nonzero differential: $8_{19} = T(3,4)$

For even n the double cover $\Sigma(2,3,n)$ carries nontrivial reducibles ($H_1 \neq 0$), so maps involving the reducible can occur in $\widetilde{C}$. Proposition 5.5 shows that its differential is nonzero precisely when $n \equiv 4 \pmod 6$. We treat the smallest case, $8_{19} = T(3,4)$, where the differential is also known geometrically on the pillowcase, in two ways. (Rank vectors (r_0, r_1, r_2, r_3) record ranks in gradings $0,1,2,3$, as in Section 5.)

6.1. **Structural derivation.** By Theorem 1.2 and Proposition 4.1, $T(3,4)$ has three irreducible traceless characters, two non-dihedral (at $\psi \approx 125.3°, 54.7°$) and one dihedral (at $90°$), together with one abelian character. Via the double cover $\Sigma(2,3,4)$, whose first homology is $\mathbb{Z}/3$: the two non-dihedral characters are the two lifts of the single Σ-irreducible; the dihedral character is the nontrivial reducible; the abelian character is Θ. Through the grading framework (an irreducible contributes four generators, two in grading μ and two in grading $\mu+1$; the nontrivial reducible two; Θ one, in grading $\sigma \bmod 4 = 2$), with $\mu_{\mathrm{irr}} = 3$ and $\mu_{\mathrm{red}} = 1$, the chain complex assembles to

$$\widetilde{C}(8_{19}) = (2,1,2,2), \qquad \chi = +1,$$

matching the instanton computation of Poudel–Saveliev [PS17, Ex. 7.9]. The rank equations for a grading-respecting differential and homology $(2,1,1,1)$ force $\operatorname{rank} \partial_3 = 1$ for $\partial_3 \colon C_3 \to C_2$, with every other graded component of the differential zero. This argument determines the rank and degree of the map, but not its source and target basis vectors; those are supplied by the geometric computation below.

6.2. **The explicit bigon.** Hedden–Herald–Kirk [HHK18, §11.6] realize exactly this. With L_0 the earring figure-eight and L_1 the perturbed character variety of the torus-knot tangle with the parameters $(3,4,3,-2)$ in the notation of [HHK18, §11], the singular unperturbed variety is shaped like the letter φ, a circle with an arc through it, whence the name "φ-curve" of [HHK18, §11.6]. It resolves under perturbation into an arc R_0 and a vertically monotone circle R_1. The seven generators are r_+ (corner, $\mu = 2$), $x_1^{\pm}$ (arc), and $x_2^{\pm}, x_3^{\pm}$ (circle); the gradings are $\mu(x_1^+) = 0$, $\mu(x_1^-) = 3$, and the circle contributes one generator in each grading. There is exactly one bigon, $x_1^- \to r_+$, running into the corner along R_0 and the earring, giving $\partial x_1^- = r_+$, a rank-one map from grading 3 to grading 2. The circle bounds no bigon. The homology is

$$H_*(\widetilde{C}(8_{19})) = (1,0,0,0)_{\mathrm{arc}} \oplus (1,1,1,1)_{\mathrm{circle}} = (2,1,1,1) \cong I^{\natural}(8_{19}),$$

of rank $5 > \det = 3$.

6.3. **Comparison of the two derivations.** The structural derivation and the explicit computation give the same graded complex and the same differential. The bigon is a corner figure-eight bigon: the earring's self-crossing at a pillowcase corner linking an interior irreducible character to the reducible locus. This is precisely the mechanism absent on two-bridge knots (Section 3.2): there the Lagrangians are straight, there is no corner bigon, and $\partial = 0$; here the single corner bigon is what makes $I^\natural(8_{19})$ exceed det. The geometric computation is due to [HHK18]; the preceding rank argument independently predicts the differential's rank and degree, while the pillowcase bigon identifies its source and target.

7. The open direction

Everything above concerns generators and gradings; the analytic heart of the knot Atiyah–Floer conjecture is the bounding cochain. On two-bridge knots it is inert (Section 3.2); off them it is genuinely nonzero, as shown for the $(4,5)$-torus knot [CHKK22] and the pretzel knot $P(-2,3,5)$ [Smi24]. Here $P(-2,3,5) = T(3,5) = 10_{124}$ is itself a member of our family, the smallest $n \equiv 5 \pmod 6$ case. Smith's nonzero pillowcase correction in that example is compatible with the rank-one differential of Proposition 5.3, but no chain-level identification is asserted here. The framed complex has a nonzero differential for $n \equiv 5$ and $n \equiv 4 \pmod 6$, while it has zero differential for $n \equiv 1$ and $n \equiv 2 \pmod 6$. Two geometric problems remain.

First, the instanton side. For every torus knot the irreducible singular instanton homology is $\mathbb{Z}_{(1)}^{\lceil -\sigma/4\rceil} \oplus \mathbb{Z}_{(3)}^{\lfloor -\sigma/4\rfloor}$ (subscripts denote the grading, superscripts the rank: the rank vector is $(0, \lceil -\sigma/4\rceil, 0, \lfloor -\sigma/4\rfloor))$ with vanishing differential [DS24]. For even n, however, $\Sigma(2,3,n)$ carries a nontrivial reducible (det $= 3$, so $H_1 = \mathbb{Z}/3$), and $\widetilde{C}$ has the shape

$$\widetilde{C}(T(3,n)) \;=\; \underbrace{\tfrac{N-1}{2}\text{ irreducibles}}_{4\text{ generators each}} \oplus \underbrace{\text{one reducible}}_{2} \oplus \Theta, \qquad \dim = 2N(3,n)+1,$$

on which maps involving the reducible can appear, exactly the corner figure-eight bigon of 8_{19} (Section 6). Proposition 5.5 settles the algebraic rank in this model: its differential vanishes for $n \equiv 2 \pmod 6$ and has total rank one for $n \equiv 4 \pmod 6$. What remains open is to identify the rank-one map geometrically throughout the latter family and to explain geometrically why all candidate maps vanish in the former.

Second, and harder, the pillowcase side: to compute the bounding-cochain-deformed pillowcase invariant and match it to $I^\natural$. A rigorous uniform computation for this torus-knot family is not presently available; relevant individual calculations include the $(4,5)$-torus knot (correct rank [CHKK22]) and $P(-2,3,5)$ (nonzero only [Smi24]). The companion [Wue26] studies finite polygon calculations for members of the pretzel family $P(-2,3,q)$ within the immersed-curve combinatorial model of [HK24, Smi24]. Establishing the full Maurer–Cartan equation, including repeated and higher insertions, and proving the analytic identification remain open. Both problems lie beyond the present, purely representation-theoretic, account.

Appendix A. Changes from version 2

This is version 3. It corrects, completes, and narrows version 2 of July 30, 2026, as follows.

1. *A false statement about even n is corrected.* Version 2 asserted that a nonzero differential appears for every even n. It does not. Proposition 5.5 proves that in the framed complex $\widetilde{C}$ the differential vanishes when $n \equiv 2 \pmod 6$ and has total rank one when $n \equiv 4 \pmod 6$; the smallest counterexample to the earlier assertion is $T(3,8)$.
2. *The grading split is proved rather than evaluated.* Version 2 obtained Proposition 5.3 by evaluating (4)–(5), using the assertion that $\rho \equiv 3$ on $\Sigma(2,3,n)$, which is not generally valid. The proposition is now proved from Daemi–Scaduto's torus-knot theorem [DS24] together with the two-lift dictionary for the branched-cover correspondence [DS24b, PS17], and the index evaluation is retained only as an independent check for all odd $n \leq 43$.
3. *The homology ranks are obtained differently.* The reduced Khovanov ranks now come from Schütz's Bar–Natan chain decomposition for three-braids [Sch25], in place of the

earlier appeal to Turner's spectral sequence and Shumakovitch's torsion theorem; the Alexander lower bound is attributed to [KM10]; and the coefficient-norm computation is carried out for every n coprime to 3, as Proposition 5.5 requires.

(4) *Theorem 1.2 has a complete proof.* Version 2 established only that a binary-dihedral traceless character forces n even. The converse construction, which produces that character for every even n, is now given. The standing hypothesis $n \geq 2$ is stated, and the integrality of ϕ_p in the proof of Theorem 1.1 is justified.

(5) *The chain-level object is named precisely.* Statements formerly made about "the reduced singular-instanton chain complex $IC^{\natural}$" are now made about the framed complex $C_* \oplus C_{*-1} \oplus \mathbb{Z}_{(0)}$, and every differential rank in this paper refers to that model.

(6) *Three claims are narrowed.* In Section 6 the rank argument is now claimed to determine only the rank and the degree of the differential; version 2 read its source and target off that argument as well, which the rank equations do not supply. The assertion that Smith's nonzero bounding cochain for $P(-2,3,5)$ *is* the differential of $\widetilde{C}$ is withdrawn: the two are compatible, and no chain-level identification is made here. References to the bounding cochains computed in the companion paper [Wue26] are rescoped to match its own version 3, in which those computations are restated as finite polygon calculations.

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